Haskell – Lab 5

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Solution for the Exercises from Chapter 7 – Higher-Order Functions

Solution for the Additional Exercises

```
squarePositives :: [Int] -> [Int]
squarePositives xs = map (^2) (filter (>= 0) xs)

squarePositives2 :: [Int] -> [Int]
squarePositives2 = map (^2) . filter (>= 0)
```

```
import Data.Char (toUpper)
capitalizeLong :: Int -> [String] -> [String]
capitalizeLong n = map (map toUpper) . filter (\x -> length x > n)
```

1. Partial functions

2. map f (filter p xs)

```
ghci>
ghci> func1 f p xs = [f x | x <- xs, p x]
ghci> func1 (+1) even [1,2,3,4,5]
[3,5]
ghci> func2 f p xs = map f (filter p xs)
ghci> func2 (+1) even [1,2,3,4,5]
[3,5]
ghci>
```

3.

```
map' :: (a -> b) -> [a] -> [b]
map' f ys = foldr (\x xs -> f x : xs) [] ys

filter' :: (a -> Bool) -> [a] -> [a]
filter' p ys = foldr (\x xs -> if p x then x : xs else xs) [] ys
```

```
ghci>
ghci> map' f ys = foldr (\x xs -> f x:xs) [] ys
ghci> map' (+1) [1,2,3,4,5]
[2,3,4,5,6]
ghci> filter' p ys = foldr (\x xs -> if p x then x:xs else xs) [] ys
ghci> filter' even [1,2,3,4,5]
[2,4]
ghci>
```

Solution for the Exercises from Chapter 8 – Declaring Types and Classes

Solution for the Additional Exercises

```
data Employee = FullTime String Float | Contractor String Float

monthlyPay :: Employee -> Float
monthlyPay (fulltimeSalary) = salary / 12
monthlyPay (contractorRate) = rate * 160
```

1.

```
data Nat = Zero | Succ Nat
                deriving Show

add' :: Nat -> Nat -> Nat
add' Zero n = n
add' (Succ m) n = Succ (add' m n)

mult' :: Nat -> Nat -> Nat
mult' Zero     n = Zero
mult' (Succ m) n = add' n (mult' m n)
```

```
ghci> add' (Succ (Succ Zero)) (Succ (Succ (Succ Zero)))
Succ (Succ (Succ (Succ (Succ Zero))))
ghci>
ghci> mult' (Succ (Succ Zero)) (Succ (Succ (Succ Zero)))
Succ (Succ (Succ (Succ (Succ (Succ Zero)))))
ghci>
```

Exercises
from
Chapter 10 – Interactive Programming

Implement the game of nim in Haskell, where the rules of the game are as follows:

- The board comprises five rows of stars:

```
1: * * * * *
2: * * * *
3: * * *
4: * *
5: *
```

- Two players take it turn about to remove one or more stars from the end of a single row.

- The winner is the player who removes the last star or stars from the board.

Hint:

Represent the board as a list of five integers that give the number of stars remaining on each row. For example, the initial board is [5,4,3,2,1].

Windows PowerShell

```
Windows PowerShell
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Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Users\User> cd Desktop\Haskell
PS C:\Users\User\Desktop\Haskell>
PS C:\Users\User\Desktop\Haskell> ghci
GHCi, version 9.4.8: https://www.haskell.org/ghc/  :? for help
ghci>
ghci> :load nim.hs
[1 of 2] Compiling Main             ( nim.hs, interpreted )
Ok, one module loaded.
ghci>
ghci> playNim
1: * * * * *
2: * * * *
3: * * *
4: * *
5: *

Player 1 turn:
Which line? 1
How many stars to remove? 5
1:
2: * * * *
3: * * *
4: * *
5: *

Player 2 turn:
Which line? 2
How many stars to remove? 4
1:
2:
3: * * *
4: * *
5: *
```

```
Windows PowerShell

Player 1 turn:
Which line? 3
How many stars to remove? 3
1:
2:
3:
4: * *
5: *

Player 2 turn:
Which line? 4
How many stars to remove? 2
1:
2:
3:
4:
5: *

Player 1 turn:
Which line? 5
How many stars to remove? 1
Player 1 wins!
ghci>
```

You can run the attached **nim.exe** (application) by typing **./nim** directly on the terminal, as follows: `PS C:\Users\User\Desktop\Haskell> ./nim`

You can start writing the program from the code skeleton called **nim_framework.hs**.

The **hangman.hs** program discussed in the lecture is also a good reference.

Exercises
from
Chapter 15 – Lazy Evaluation

1. Define a program

```
fibs :: [Integer]
```

that generates the infinite Fibonacci sequence

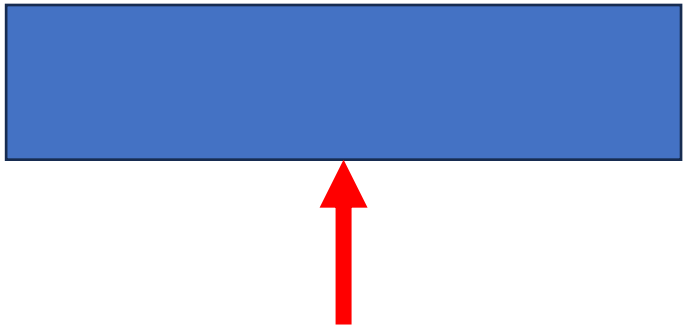
```
[0,1,1,2,3,5,8,13,21,34…
```

Using the following simple procedure:

a) The first two numbers are 0 and 1.
b) The next is the sum of the previous two.
c) Return to step b)

```
ghci>
ghci> fibs
[0,1,1,2,3,5,8,13,21,34,55,89,144,233,377,610,987,1597,2584,4181,6765,10946,17711,28657,46368,75025,121393,196418,317811,514229,832040,1346269,2178309,3524578,5702887,9227465,14930352,24157817,39088169,63245986,102334155,165580141,267914296,433494437,701408733,1134903170,1836311903,2971215073,4807526976,7778742049,12586269025,20365011074,32951280099,53316291173,86267571272,139583862445,225851433717,365435296162,591286729879,95672202
```

```
fibs :: [Integer]
fibs = 0:fibs2 0 1
  where
    fibs2 :: Integer -> Integer -> [Integer]
    fibs2 a b = b:
```



Recursion of fibs2

2. Define a function

```
fib :: Integer -> Integer
```

that calculates the nth Fibonnaci number.

```
ghci>
ghci> fib 7
8
ghci> fib 9
21
ghci> fib 10
34
ghci>
```

```
fib :: Integer -> Integer
fib n = fibs !! (fromInteger [redacted])
```

nth Fibonnaci number

We use fromInteger to convert its argument (Integer type) into Int type.

This is because the !! operator requires a list at its left and an Int (not Integer) at its right.